SKILL-2
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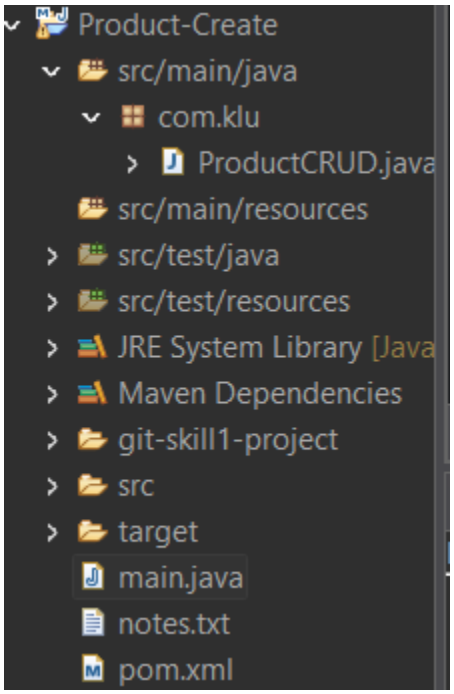
SECTION:59
ID:2400030211



ProductCRUD.java:

```
package com.klu;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class ProductCRUD {

    static String url = "jdbc:mysql://localhost:3306/product";
```

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```
static String user = "root";
static String password = "root";

static Scanner sc = new Scanner(System.in);

public static void main(String[] args) {
        createTableIfNotExists();

    int choice;

    do {
        System.out.println("\n----- PRODUCT CRUD MENU ----- ");
        System.out.println("1. Insert Product");
        System.out.println("2. View Products");
        System.out.println("3. Update Price");
        System.out.println("4. Update Quantity");
        System.out.println("5. Delete Product");
        System.out.println("6. Exit");
        System.out.print("Enter choice: ");

        choice = sc.nextInt();

        switch (choice) {
            case 1:
                insertProduct();
                break;
            case 2:
                viewProducts();
                break;
            case 3:
                updatePrice();
```

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```
            break;
        case 4:
            updateQuantity();
            break;
        case 5:
            deleteProduct();
            break;
        case 6:
            System.out.println("Exiting...");
            break;
        default:
            System.out.println("Invalid choice!");
    }
  } while (choice != 6);
}
static void createTableIfNotExists() {
  try (Connection con = DriverManager.getConnection(url, user, password)) {
    String sql = "CREATE TABLE IF NOT EXISTS product (" +
                "product_id INT PRIMARY KEY," +
                "product_name VARCHAR(50)," +
                "product_descript VARCHAR(100)," +
                "price DOUBLE," +
                "product_quantity INT)";
    con.createStatement().executeUpdate(sql);
  } catch (Exception e) {
    e.printStackTrace();
  }
}

// INSERT
static void insertProduct() {
```

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```
    try (Connection con = DriverManager.getConnection(url, user, password)) {

        String sql = "INSERT INTO product VALUES (?, ?, ?, ?, ?)";
        PreparedStatement ps = con.prepareStatement(sql);

        System.out.print("Enter Product ID: ");
        ps.setInt(1, sc.nextInt());

        System.out.print("Enter Product Name: ");
        ps.setString(2, sc.next());

        System.out.print("Enter Description: ");
        ps.setString(3, sc.next());

        System.out.print("Enter Price: ");
        ps.setDouble(4, sc.nextDouble());

        System.out.print("Enter Quantity: ");
        ps.setInt(5, sc.nextInt());

        ps.executeUpdate();
        System.out.println(" Product Inserted Successfully");

    } catch (Exception e) {
        e.printStackTrace();
    }
}

// READ
static void viewProducts() {
    try (Connection con = DriverManager.getConnection(url, user, password)) {
```

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```
        String sql = "SELECT * FROM product";
        PreparedStatement ps = con.prepareStatement(sql);
        ResultSet rs = ps.executeQuery();

        System.out.println("\nID  Name  Description  Price  Quantity");

        while (rs.next()) {
            System.out.println(
                    rs.getInt("product_id") + "  " +
                    rs.getString("product_name") + "  " +
                    rs.getString("product_descript") + "  " +
                    rs.getDouble("price") + "  " +
                    rs.getInt("product_quantity")
            );
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
}

// UPDATE PRICE
static void updatePrice() {
    try (Connection con = DriverManager.getConnection(url, user, password)) {

        String sql = "UPDATE product SET price = ? WHERE product_id = ?";
        PreparedStatement ps = con.prepareStatement(sql);

        System.out.print("Enter New Price: ");
        ps.setDouble(1, sc.nextDouble());
```

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```
        System.out.print("Enter Product ID: ");
        ps.setInt(2, sc.nextInt());

        ps.executeUpdate();
        System.out.println(" Price Updated Successfully");

    } catch (Exception e) {
        e.printStackTrace();
    }
}

// UPDATE QUANTITY
static void updateQuantity() {
    try (Connection con = DriverManager.getConnection(url, user, password)) {

        String sql = "UPDATE product SET product_quantity = ? WHERE product_id = ?";
        PreparedStatement ps = con.prepareStatement(sql);

        System.out.print("Enter New Quantity: ");
        ps.setInt(1, sc.nextInt());

        System.out.print("Enter Product ID: ");
        ps.setInt(2, sc.nextInt());

        ps.executeUpdate();
        System.out.println(" Quantity Updated Successfully");

    } catch (Exception e) {
        e.printStackTrace();
    }
```

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```
    }

    // DELETE
    static void deleteProduct() {
        try (Connection con = DriverManager.getConnection(url, user, password)) {

            String sql = "DELETE FROM product WHERE product_id = ?";
            PreparedStatement ps = con.prepareStatement(sql);

            System.out.print("Enter Product ID: ");
            ps.setInt(1, sc.nextInt());

            ps.executeUpdate();
            System.out.println(" Product Deleted Successfully");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.klu</groupId>
  <artifactId>Product-Create</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <dependencies>
    <dependency>
```

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```
    <groupId>mysql</groupId>
    <artifactId>mysql-connector-java</artifactId>
    <version>8.0.33</version>
  </dependency>

 </dependencies>
</project>
```

OUTPUT :

```
----- PRODUCT CRUD MENU -----
1. Insert Product
2. View Products
3. Update Price
4. Update Quantity
5. Delete Product
6. Exit
Enter choice: 1
Enter Product ID: 231
Enter Product Name: apple
Enter Description: vry good
Enter Price: 200
Enter Quantity: 1
☑ Product Inserted Successfully
```

Viva Questions:

1. What is ORM and why is it used in application development?

ORM (Object Relational Mapping) maps Java objects to database tables and is used to reduce SQL code and simplify database operations.

2. Which annotation is used to mark a class as a JPA entity?

The @Entity annotation is used to mark a class as a JPA entity.

3. Why must every Hibernate entity have a primary key?

A primary key is required to uniquely identify each entity record in the database.

4. How do you update an existing record using Hibernate?

Update the entity object and call update() or saveOrUpdate() within a transaction.

5. What is the difference between get() and load() in Hibernate?

get() returns the object immediately or null, while load() returns a

proxy and throws an exception if not found.

Git Hub Link: https://github.com/hemalatharachakonda/2400030211-Skill_experiments.git

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